17/11/2025

Tutorial

C.T. F.S } for periodic signals
D.T. F.S }

C.T. F.T } for aperiodic signals
D.T. F.T } → Later included periodic signals

C.T → Laplace
D.T → Z-transform } if my C.T. signal is not absolutely integrable → C.T. F.T. does not exists
→ How to analyze → use Laplace

If our D.T. signal is not absolutely summable
D.T. F.T. does not exist
→ Z-transform

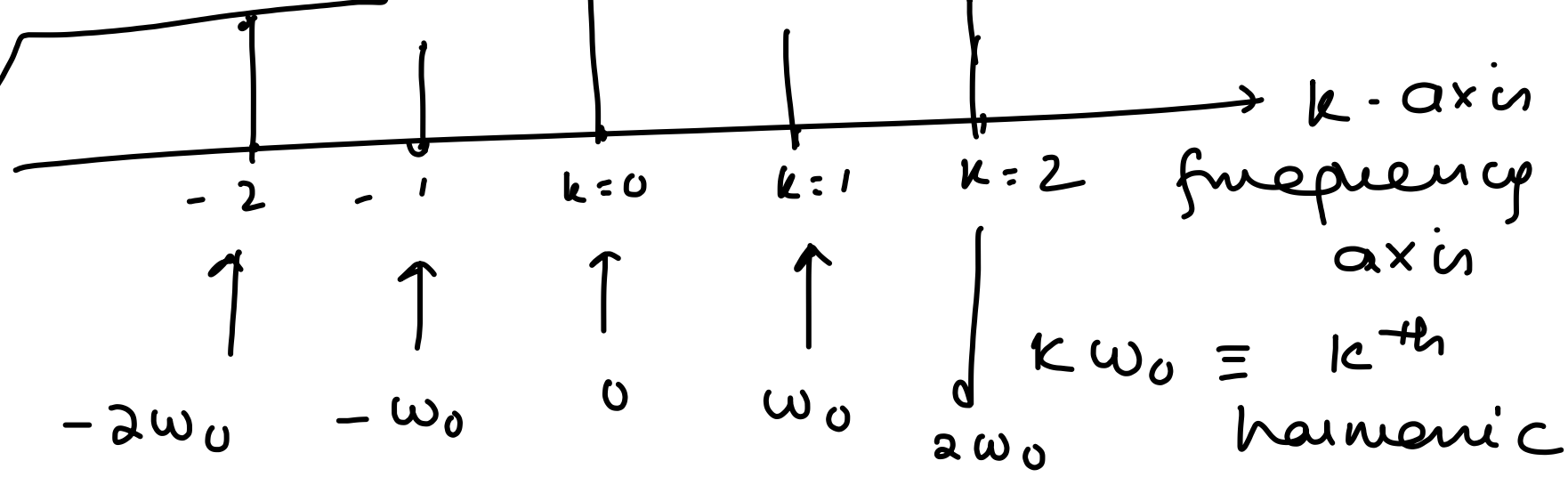
C.T.F.S

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$ ← Synthesis eqn.

$\omega_0 = \frac{2\pi}{T}$

T → fundamental time period.

$$a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t} dt$$ ← analysis eqn.



line Spectrum ⟹

D.T.F.S. Synthesis eq.:

$$x[n] = \sum_{\langle k=N \rangle} a_k \frac{e^{jk\frac{2\pi}{N}n}}{} = \sum_{k=\langle N \rangle} a_k \phi_k[n]$$

∵ in frequency domain, for discrete-time signals, frequencies are unique in 2π interval.

$k = 0 \longrightarrow \phi_0[n] = e^{j0.n} = 1$

$k = 1 \longrightarrow \phi_1[n] = e^{j\left(\frac{2\pi}{N}\right)n}$

$k = 2 \longrightarrow \phi_2[n] = e^{j\left(\frac{2\pi}{N}.2\right)n}$

$k = N-1 \longrightarrow \phi_{N-1}[n] = e^{j\left(\frac{2\pi\,(N-1)}{N}\right)n}$

$k = N \longrightarrow \phi_N[n] = e^{j\frac{2\pi}{N}.Nn} = e^{j2\pi n} = 1 = e^{j0} = \phi_0[n]$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] \, e^{-jk\frac{2\pi}{N}n}$$

C.T.F.T.

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \, e^{j\omega t} \, d\omega \quad \leftarrow \text{Synthesis eq.}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \, e^{-j\omega t} \, dt$$

D.T.F.T.

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

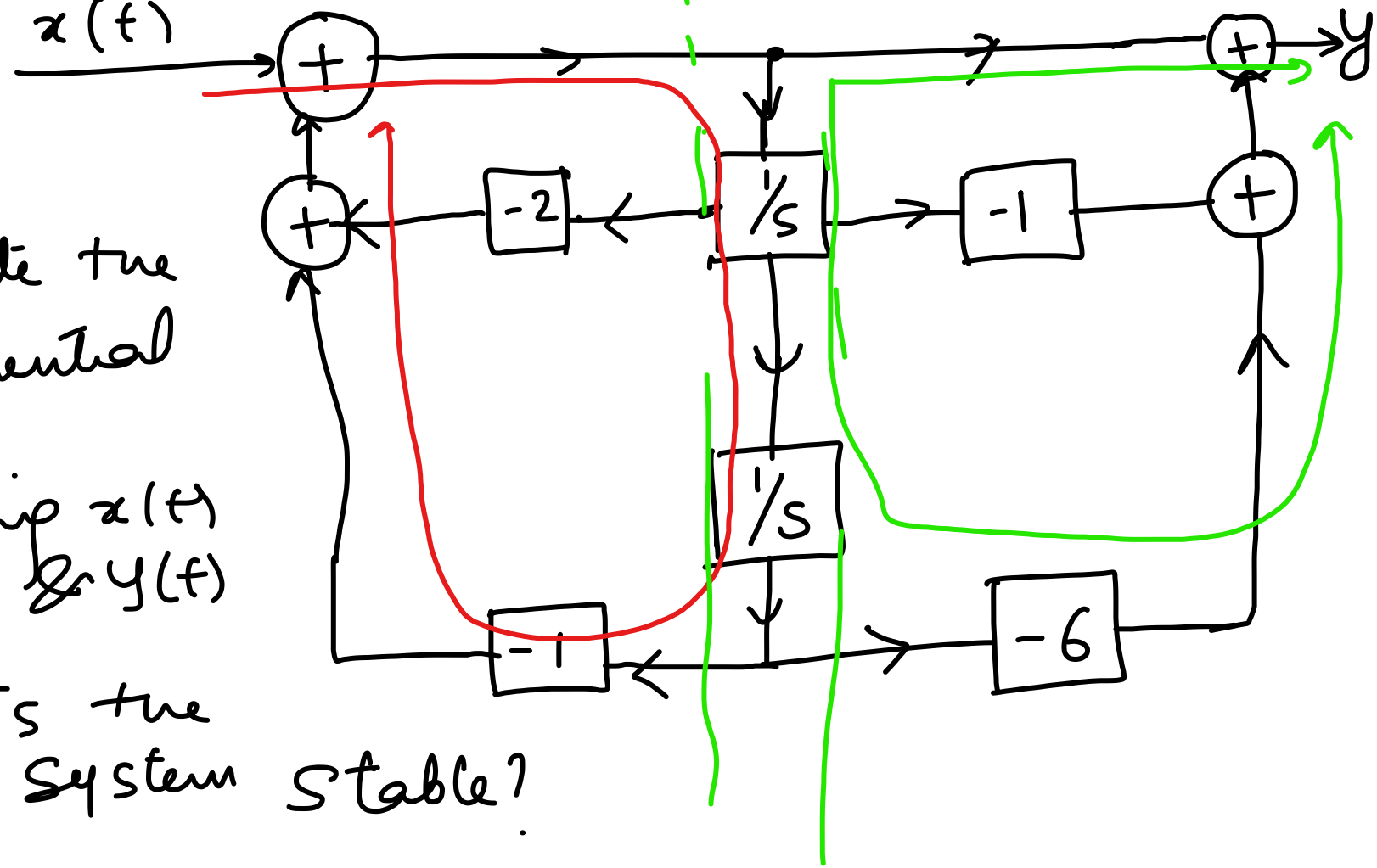
Laplace

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

Specify ROC ≡ Region of convergence ⟹ those values of s for which X(s) does not go to infinity

or $X(s)$ is finite.

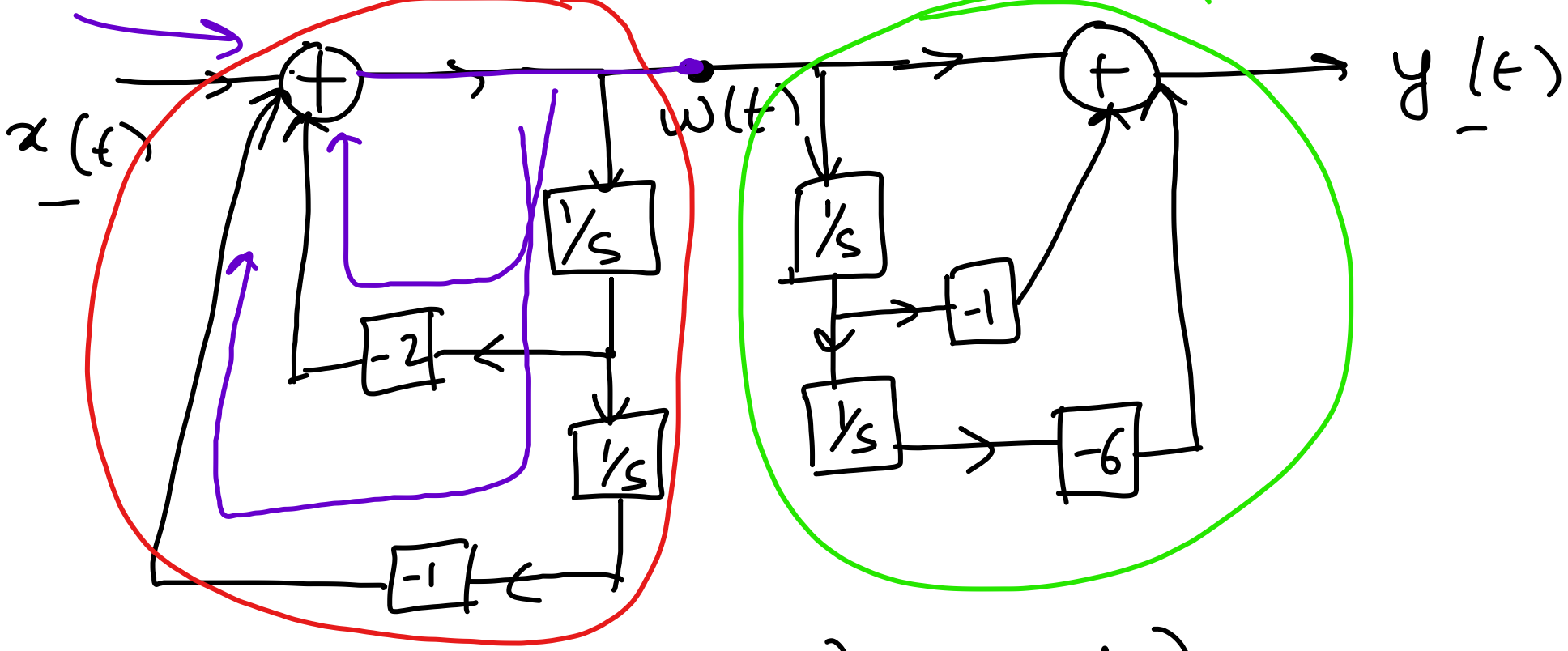
9.35



(a) Write the differential eqn. relating $x(t)$ & $y(t)$

(b) Is the system stable?

Given the system is causal LTI



$$\frac{Y(s)}{X(s)}$$

$$X(s) - 2\frac{W(s)}{s} - \frac{W(s)}{s^2} = W(s)$$

$$X(s) = W(s) + 2\frac{W(s)}{s} + \frac{W(s)}{s^2}$$

$$s^2 X(s) = s^2 W(s) + 2sW(s) + W(s)$$

$$\frac{W(s)}{X(s)} = \frac{s^2}{s^2 + 2s + 1} = \frac{s^2}{(s+1)^2}$$

$$Y(s) = W(s) - \frac{W(s)}{s} - \frac{6\,W(s)}{s^2}$$

$$\frac{Y(s)}{W(s)} = \frac{s^2 - s - 6}{s^2}$$

$$\frac{Y(s)}{X(s)} = \frac{Y(s)}{W(s)} \times \frac{W(s)}{X(s)} = \frac{s^2 - s - 6}{s^2 + 2s + 1}$$

$$(s^2 + 2s + 1)\,Y(s) = (s^2 - s - 6)\,X(s)$$

$$\frac{d^2}{dt^2}y(t) + 2\frac{dy(t)}{dt} + y(t) = \frac{d^2}{dt^2}x(t) - \frac{d}{dt}x(t) - 6x(t)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{(s-3)(s+2)}{(s+1)^2}$$

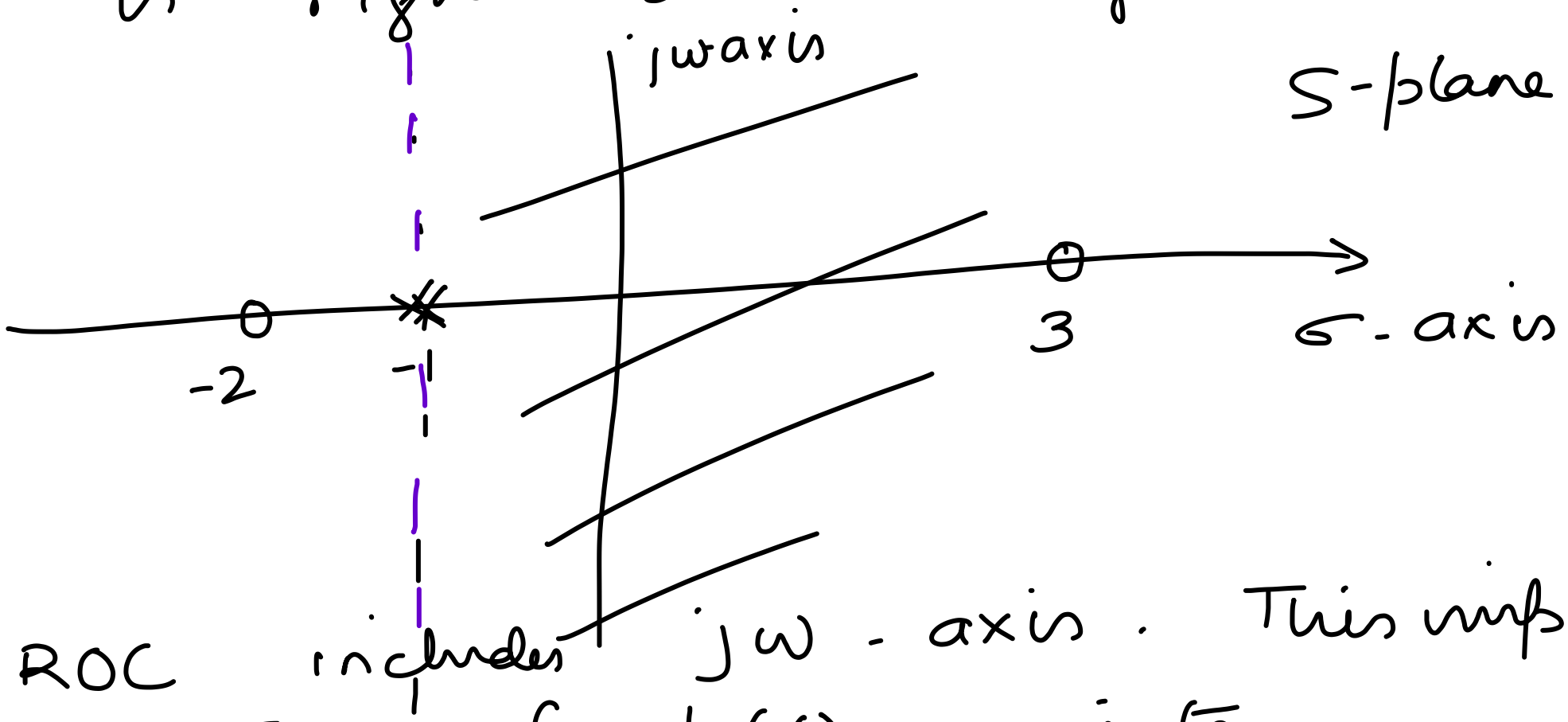
zeros are at $s = 3$ & $s = -2$

poles at $s = -1$ (order 2)

$H(s)$ = rational in s

$= \dfrac{N(s)}{D(s)} =$ Numerator is a polynomial in s / Denominator is a polynomial in s

if H(s) is rational in s & the system is causal, then ROC is right to the rightmost pole.



∴ ROC includes jω - axis. This implies that F.T. of h(t) exists.

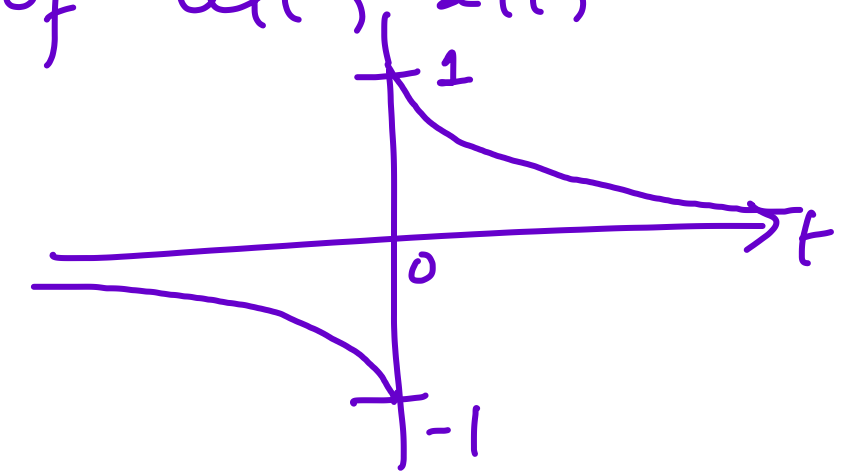
This implies that $h(f)$ is absolutely integrable $\int_{-\infty}^{\infty} |h(f)|\,df < \infty$

But, this is the condition that $h(f)$ should satisfy for BIBO stability

$\therefore$ this implies that $h(f)$ is stable.

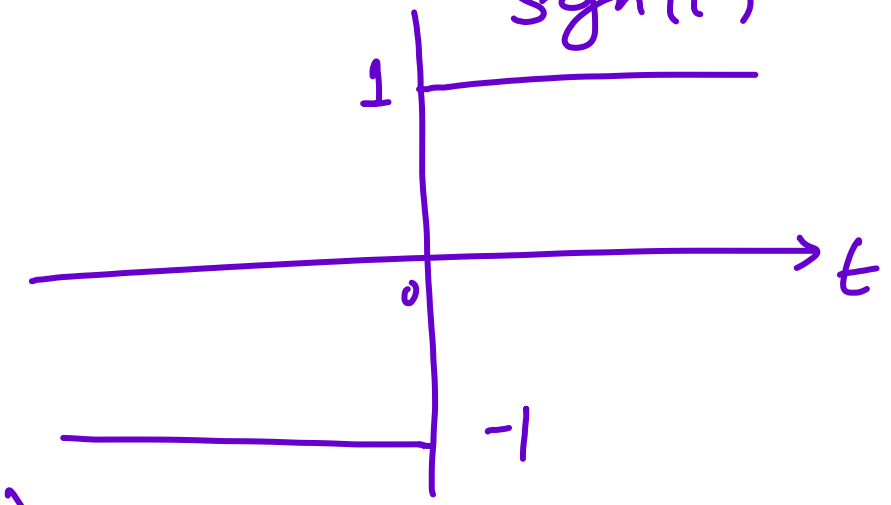
Derivation of F.T. of u(t)

$$x(t) = \begin{cases} e^{-at} & t > 0 \\ 0 & t = 0 \\ -e^{at} & t < 0 \end{cases}$$



$$u(t) = \frac{1}{2}\left(Sgn(t) + 1\right) \quad \text{——— (A)}$$

$a \to 0$



① First compute F.T. of $x(t)$.

② Then put limit $a \to 0$ & compute F.T. of $Sgn(t)$

③ Compute F.T. of $u(t)$ using (2) & (A) above.

① F.T. of $x(t)$

$$X(j\omega) = \int_0^{\infty} e^{-at} \cdot e^{-j\omega t}\, dt - \int_{-\infty}^{0} e^{at} \cdot e^{-j\omega t}\, dt$$

$$= \frac{e^{-(a+j\omega)t}}{-(a+j\omega)}\Bigg|_0^{\infty} - \frac{e^{(a-j\omega)t}}{a-j\omega}\Bigg|_{-\infty}^{0}$$

$$= \frac{1}{a+j\omega} - \frac{1}{a-j\omega} = \frac{a-j\omega-a-j\omega}{a^2-j^2\omega^2}$$

$$= \frac{-2j\omega}{a^2+\omega^2}$$

$\therefore \boxed{j^2 = -1}$

$$x(j\omega) = \frac{-2j\omega}{a^2+\omega^2}$$

$x(t) \longrightarrow Sgn(t)$ if $a \longrightarrow 0$

& $x(j\omega)\Big|_{a\to 0} \longrightarrow \frac{-2j\omega}{\omega^2} = \frac{+2j}{j^2\omega}$

$$= \frac{2}{j\omega}$$

$$x(t) \rightleftharpoons \frac{-2j\omega}{a^2+\omega^2}$$

$$Sgn(t) \rightleftharpoons \frac{2}{j\omega}$$

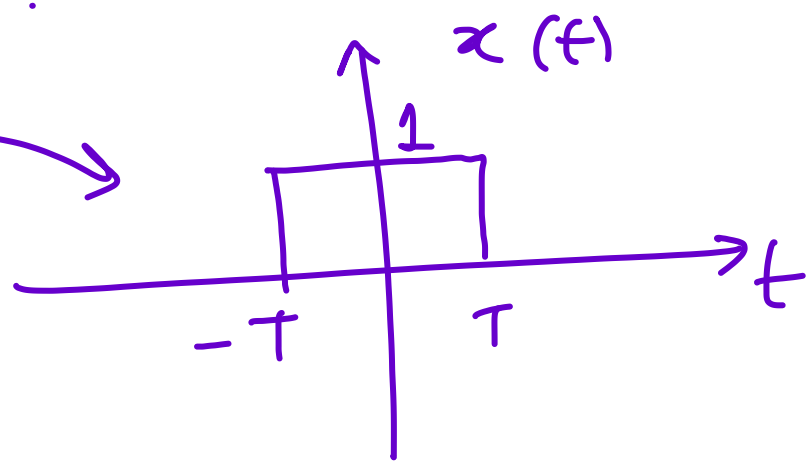
$$1 \overset{F.T}{\rightleftharpoons} 2\pi\delta(\omega)$$

$$(\mathrm{Sgn}(t)+1) \rightleftharpoons \frac{2}{j\omega} + 2\pi\delta(\omega)$$

$$u(t) = \frac{1}{2}(\mathrm{Sgn}(t)+1) \rightleftharpoons U(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

Duality in F.T.

① $x(t) = \mathrm{rect}\left(\frac{t}{T}\right)$



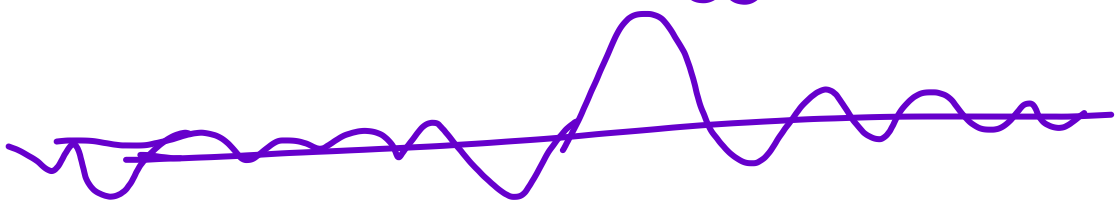
$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$= \int_{-T}^{T} e^{-j\omega t}\, dt = \frac{e^{-j\omega t}}{(-j\omega)} \Bigg|_{-T}^{T}$$

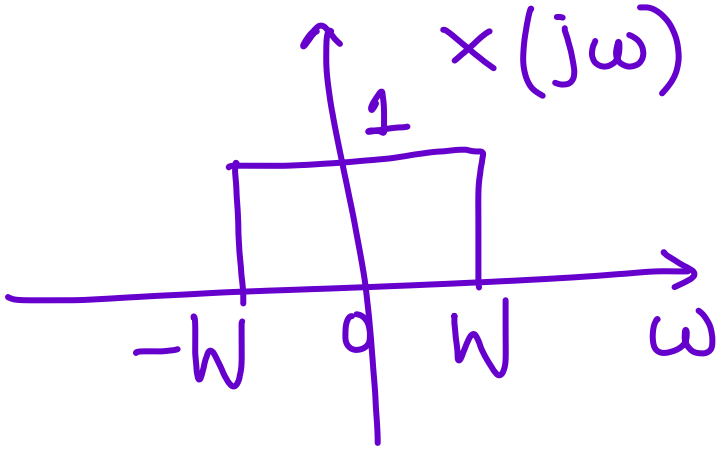
$$= \frac{e^{-j\omega T} - e^{j\omega T}}{-j\omega} = \frac{2\left(e^{j\omega T} - e^{-j\omega T}\right)}{\omega\,(2j)}$$

$$= \frac{2\sin\omega T}{\omega}$$

$$\text{rect}\left(\frac{t}{T}\right) \Longleftrightarrow \frac{2\sin\omega T}{\omega}$$



(2) $X(j\omega) = \text{rect}\left(\frac{\omega}{W}\right)$



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

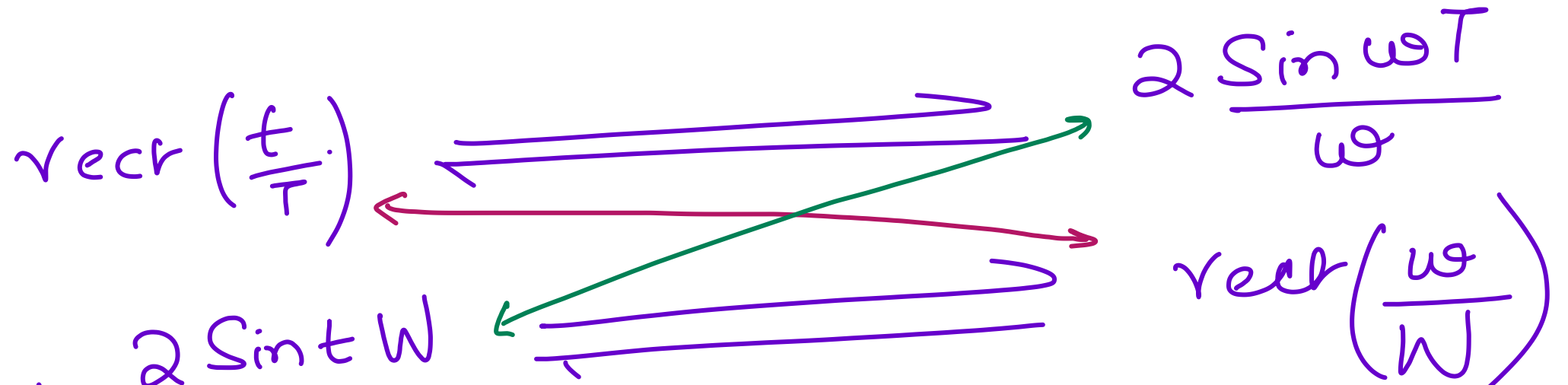
$$= \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t}\, d\omega = \frac{1}{2\pi}\left.\frac{e^{j\omega t}}{(jt)}\right|_{-W}^{W}$$

$$= \frac{2}{2\pi}\left(\frac{e^{jtW} - e^{-jtW}}{2jt}\right)$$

$$= \frac{2}{2\pi} \frac{\sin tW}{t}$$

$$= \frac{1}{2\pi} \cdot \left(\frac{2 \sin tW}{t} \right)$$

$$\frac{1}{2\pi} \cdot \frac{2 \sin tW}{t}$$



=

$\text{rect}\left(\frac{t}{T}\right) \rightleftharpoons 2\frac{\sin \omega T}{\omega}$ ✓

time domain to frequency domain.

$\frac{2\sin(tW)}{t} \rightleftharpoons ?$

$2\pi \,\text{rect}\left(\frac{\omega}{W}\right)$

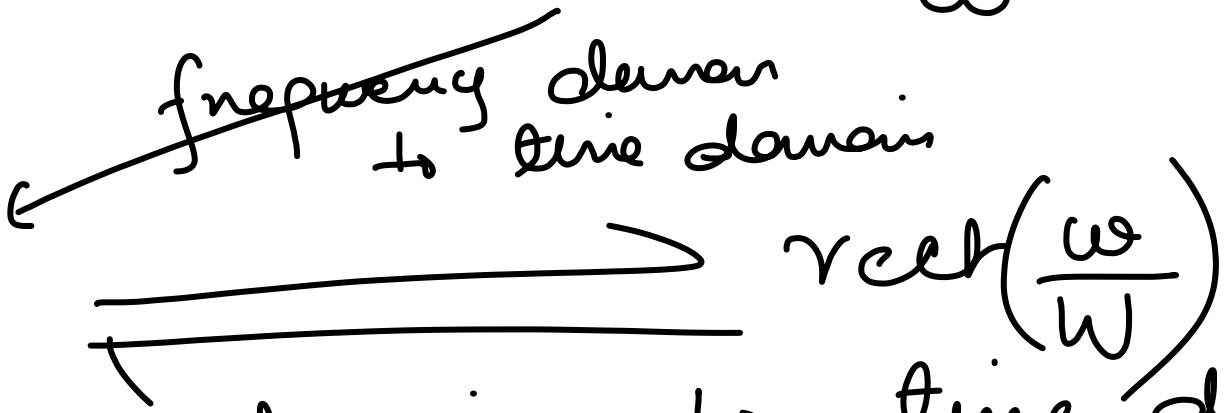
time domain to frequency domain

change $t \longrightarrow -\omega$

& multiply by 2π

$\frac{1}{2\pi} \cdot \frac{2\sin tW}{t} \rightleftharpoons \text{rect}\left(\frac{\omega}{W}\right)$

$$rect\left(\frac{t}{T}\right) \rightleftharpoons \frac{2\sin\omega T}{\omega}$$



from frequency domain to time domain

change $\omega \longrightarrow (-t)$

& divide by 2π

$$\frac{1}{2\pi}\, 2\, \frac{\sin(-t)W}{-t} = \frac{1}{2\pi}\, \frac{2\sin tW}{t}$$

Ex. Find F.T. of $x(t) = \frac{2}{1+t^2}$

$X(j\omega) = ?$

$y(t) = e^{-a|t|} \quad a > 0$

$$y(t) = \begin{cases} e^{-at} & t \geq 0 \\ e^{at} & t < 0 \end{cases}$$

$$Y(j\omega) = \int_0^\infty e^{-at} \cdot e^{-j\omega t}\, dt + \int_{-\infty}^0 e^{at} e^{-j\omega t}\, dt$$

$$= \frac{e^{-(a+j\omega)t}}{-(a+j\omega)}\Bigg|_0^\infty + \frac{e^{(a-j\omega)t}}{(a-j\omega)}\Bigg|_{-\infty}^0$$

$$Y(j\omega) = \frac{1}{a+j\omega} + \frac{1}{a-j\omega} = \frac{2a}{a^2+\omega^2}$$

Let $a = 1$

$$y(t) = e^{-|t|}$$

$$Y(j\omega) = \frac{2}{1+\omega^2}$$

$e^{-|t|}$ $\rightleftharpoons$ $\frac{2}{1+\omega^2}$

time to frequency

$\frac{2}{1+t^2}$ $\rightleftharpoons$?

change t to $(-\omega)$

& multiply by 2π

$e^{-|t|}$ $\rightleftharpoons$ $\frac{2}{1+\omega^2}$

$\frac{2}{1+t^2}$ $\rightleftharpoons$ $2\pi e^{-|\omega|}$

$$e^{-|t|} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{2}{1+\omega^2}\right)e^{+j\omega t}\,d\omega$$

$$y(t) = \frac{1}{2\pi}\int Y(j\omega)\,e^{j\omega t}\,d\omega$$

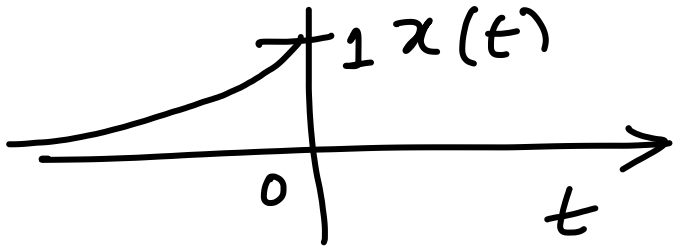
$$e^{-|\omega|} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{2}{1+t^2}\,e^{-j\omega t}\,dt$$

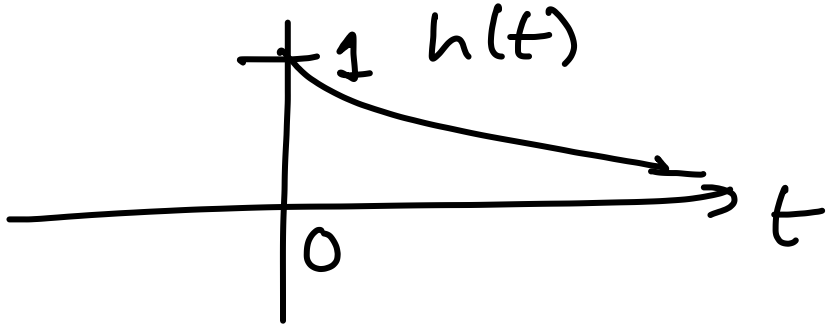
$$2\pi\, e^{-|\omega|} = \int_{-\infty}^{\infty} \left(\frac{2}{1+t^2}\right) e^{-j\omega t}\, dt$$

$$= F.T.\left(\frac{2}{1+t^2}\right)$$

C.T. Convolution $a > 0$

$$x(t) = e^{at}\, u(-t), \quad h(t) = e^{-at}\, u(t)$$





$$y(t) = x(t) * h(t)$$

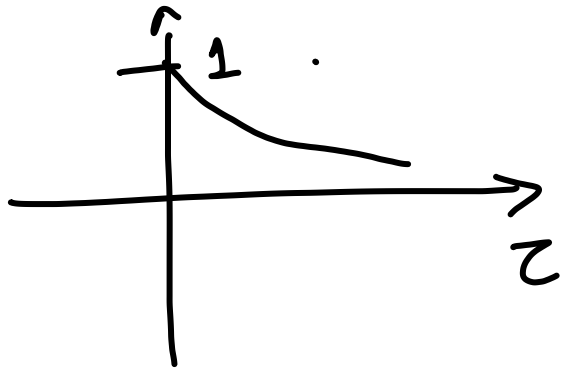
$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau = \int_{-\infty}^{\infty} x(t)\, h(t-\tau)\, d\tau$$

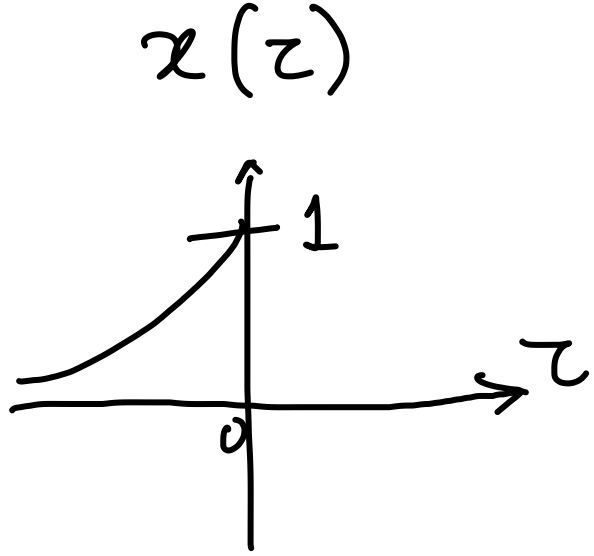
we will flip $x(\tau)$ around vertical axis & then Shift from $t = -\infty$ to $t = \infty$ (Left to right)

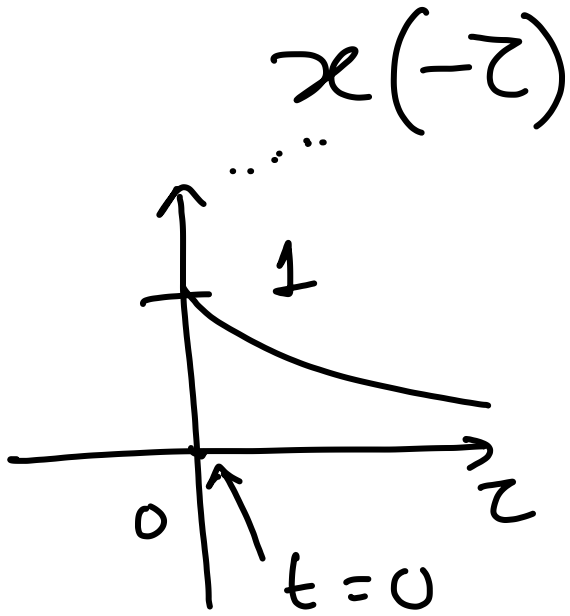
we will flip $h(\tau)$ about vertical y-axis & then shift it from $t = -\infty$ to $t = \infty$ (Left to right)

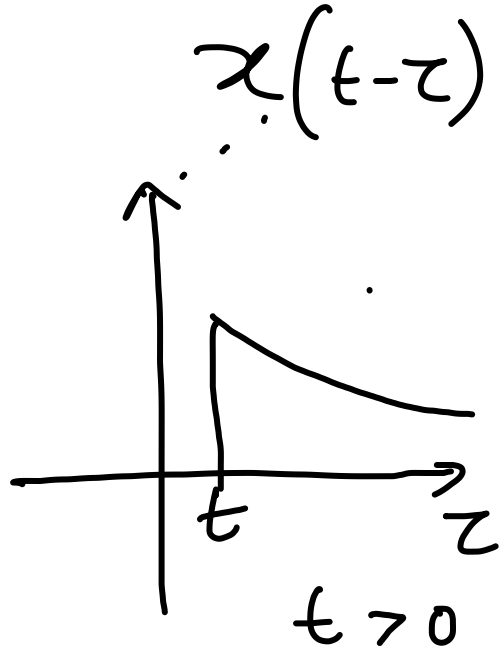
We choose to flip $x(t)$

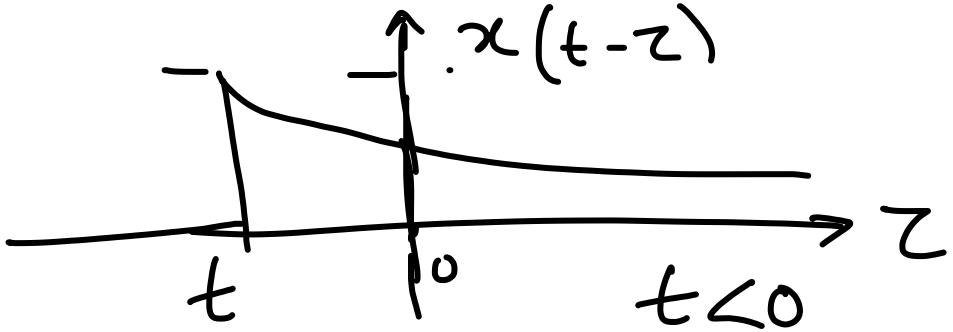
$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$











$$y(t) = \int_t^{\infty} e^{-a\tau}\, e^{+a(t-\tau)}\, d\tau \qquad t > 0$$

$$y(t) = \int_0^{\infty} e^{-a\tau}\, e^{+a(t-\tau)}\, d\tau \qquad t < 0$$

$\underline{t > 0}$

$$y(t) = e^{at} \int_t^{\infty} e^{-2a\tau}\, d\tau$$

$$= e^{at} \cdot \frac{e^{-2a\tau}}{-2a} \Bigg|_t^{\infty}$$

$$y(t) = \frac{e^{at} \cdot e^{-2at}}{2a} = \frac{e^{-at}}{2a} \qquad t > 0$$

for $t < 0$

$$y(t) = \int_0^{\infty} e^{-a\tau} \, e^{a(t-\tau)} \, d\tau$$

$$= e^{at} \frac{e^{-2a\tau}}{-2a} \Bigg|_0^{\infty}$$

$$y(t) = \begin{cases} \frac{e^{at}}{2a} & t \geq 0 \\ \frac{e^{-at}}{2a} & t < 0 \end{cases}$$

$$\int_{-\infty}^{t} x(\tau)d\tau = u(t) * x(t) \rightleftharpoons U(s) \cdot X(s)$$

$$\frac{1}{s} X(s)$$